

A Hierarchical Framework for Inverse Problems in Biological Cybernetics: Opportunities and Limitations[★]

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ABSTRACT

We consider the problem of reconstructing biological mechanisms from experimental data by means of mathematical models. This class of problems is usually known as *inverse problems*. We argue that this problem admits a natural hierarchy of increasing ambition and complexity: from estimating parameters of a known model, to discovering model structure from data, to inferring the optimality principles that govern system behavior, and finally to characterizing the strategic objectives of multiple competing biological agents. In this study we examine the mathematical foundations, practical utility, and fundamental limitations of each level of this hierarchy. Our discussion draws on arguments from both proponents and critics of the framework. For that, we assess the evidential basis for applying optimality principles to biological systems, contributing to the long-standing debate on this topic. We also discuss the relationship between this hierarchy and analogous problems in modern machine learning and artificial intelligence. Our conclusion is that the lower levels rest on quite mature theoretical and computational foundations, while the upper levels represent a compelling but more speculative research frontier whose biological validity must be established empirically rather than assumed on theoretical grounds. Throughout, we argue that the appropriate standard for optimality-based modeling in biology is not proof of correctness, but rather its potential usefulness: the generation of falsifiable hypotheses, the prediction of experimental outcomes, and the organization of otherwise disconnected observations into coherent mechanistic models.

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1. Introduction

A central goal of computational systems biology is to infer unknown biological mechanisms from observable data, a task known as an inverse problem (Engl et al., 2009; Clermont and Zenker, 2015; Banga and Villaverde, 2025). This inverse perspective also belongs to the broader tradition of biological cybernetics, which seeks to understand how living systems regulate their behavior through communication and feedback control (Wiener, 1948; Wolkenhauer, 2001; van Hemmen, 2021; Bin et al., 2022). At its most elementary level, this means estimating the kinetic parameters of a known reaction network. At its most ambitious, it means inferring the evolutionary objectives that explain why a biological system behaves the way it does, or characterizing the strategic logic underlying the interactions of competing biological agents.

Here we present and examine a four-level hierarchy of such inverse problems, assessing each level from multiple perspectives: the mathematical foundations that underpin it, the biological evidence that motivates it, and the objections that must be confronted before it can be applied with confidence. The four levels are:

1. **Parametric identification (model calibration):** Given a known model structure, estimate the parameter vector from experimental time-series data.
2. **Model discovery:** Simultaneously infer the model structure and its parameters from data.
3. **Inverse optimal control (IOC):** Infer the optimality principle (the objective function) that a biological system appears to be optimizing, given observations of its trajectories.
4. **Inverse differential games (IDG):** Recover the individual objectives of multiple competing biological agents from observations of their joint behavior.

Each level inherits the computational and identifiability challenges of all previous levels and adds qualitatively new ones. The lower levels (1 and 2) are relatively well-established. The upper levels (3 and 4) are subjects of

active debate: they rest on empirically testable (but not yet universally accepted) assumptions about the nature of biological organization, and they face identifiability challenges for which complete theoretical solutions do not yet exist.

The structure of this paper reflects this asymmetry. Sections 2 and 3, which describe the first two levels, treat the established foundations of the hierarchy. Sections 4 and 5 describe the third and fourth levels, showing more directly the controversial issues and presenting supporting and critical perspectives side by side. Next, Section 6 illustrates the hierarchy by applying it to a simple case study, a two-strain cross-feeding microbial community. The remaining three sections focus on conceptual matters. Section 7 addresses the broader question of whether optimality principles are an appropriate modeling paradigm for biology. Section 8 draws parallels with the analogous hierarchy in machine learning and artificial intelligence. Lastly, Section 9 summarizes the main conclusions and outlines a research agenda. Additionally, the Appendix provides further details about solution methods and code implementation of the case study.

2. Level 1: Parametric Identification

2.1. Mathematical foundations

Parametric identification is arguably the most classical form of biological inverse problem (Villaverde et al., 2022). Given a state-space model:

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}), \quad \mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}) \quad (1)$$

and experimental observations $\tilde{y}_{j,k,i}$ collected across conditions k , observables j , and time points t_i , the task is to find the parameter vector $\hat{\boldsymbol{\theta}}$ minimizing the weighted least-squares objective:

$$J(\boldsymbol{\theta}) = \sum_{k,j,i} \omega_{j,k,i} (\tilde{y}_{j,k,i} - y_{j,k,i}(\boldsymbol{\theta}))^2 \quad (2)$$

subject to dynamic constraints, box constraints on parameters, and any additional path or point constraints arising from the biology.

The key theoretical challenges are well understood. *Structural identifiability* asks whether the parameters can in principle be recovered from perfect, noise-free data (Anstett-Collin et al., 2020); it is formally equivalent to the observability of an augmented state-space system obtained by appending $\dot{\boldsymbol{\theta}} = \mathbf{0}$ to the dynamics (Tunali and Tarn, 1987). A parameter is structurally identifiable if and only if the corresponding component of the augmented state is observable, which can be assessed via the rank of the nonlinear observability matrix using a differential geometry approach (Hermann and Krener, 1977; Villaverde et al., 2016). Other approaches are based on differential algebra (Ljung and Glad, 1994; Hong et al., 2020), generating series (Ligon et al., 2018), or sensitivity matrices (Van Willigenburg et al., 2022), among others. *Practical identifiability* asks whether the parameters can be reliably estimated given finite, noisy data. Practical identifiability depends on the design of the

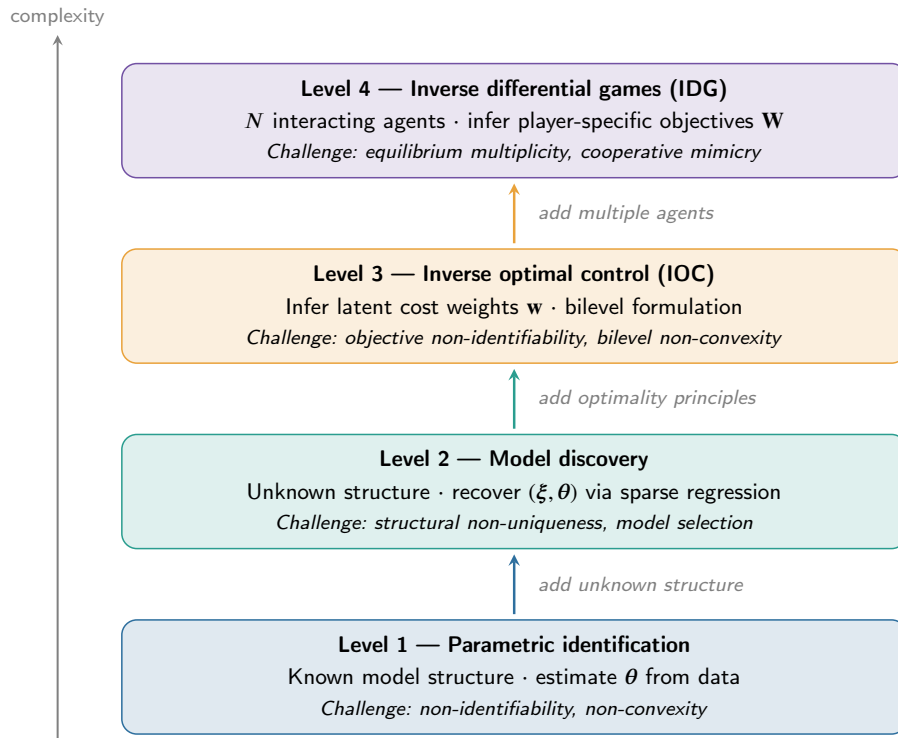


Figure 1: Hierarchical framework for inverse problems in systems biology. Each level inherits all challenges from the levels below and introduces new sources of ill-posedness and computational difficulty.

experiments and is typically quantified based on the conditioning of the Fisher Information Matrix (Lam et al., 2022; Balsa-Canto et al., 2008). Other common techniques for its analysis include the profile likelihood (Raue et al., 2009), Monte Carlo simulations (Miao et al., 2011), and Bayesian approaches (Hines et al., 2014).

2.2. Assessment

Supporting view

Level 1 is the most mature component of the hierarchy. The theoretical framework (including structural and practical identifiability analysis, robust optimization methods, and uncertainty quantification) is well developed and has been validated in numerous biological applications (Liepe et al., 2014; Villaverde et al., 2022; Banga and Villaverde, 2025; Qiao et al., 2025). The equivalence between structural identifiability and observability provides a principled diagnostic for ill-posedness before data collection. Software tools for assessing and improving parameter identifiability are available (Rey Barreiro and Villaverde, 2023; Lam et al., 2022).

Critical view

Even at Level 1, the non-convexity of the optimization landscape routinely traps gradient-based methods in local minima (Banga, 2008). For large models

with hundreds of parameters, global optimization remains computationally expensive. Furthermore, identifiability analysis is still a challenging problem; even with recent methodological advances, the computational effort demanded may be very high or even impractical for large models (Rey Barreiro and Villaverde, 2023). Many methods for analysing structural identifiability (differential algebra, Lie derivatives, generating series) work well on models of moderate size but scale poorly. Likewise, determining practical identifiability by computing profile likelihoods or sampling-based confidence intervals can be costly. Parameter non-identifiability (the existence of multiple parameter vectors producing indistinguishable outputs) is the rule rather than the exception in mechanistic biological models, and cannot be resolved by additional data alone when it is structural in origin. Current challenges include finding biologically meaningful and experimentally feasible reparameterizations, and extending the applicability of current techniques beyond ODE models (Wieland et al., 2021).

Synthesis

Level 1 is the appropriate starting point for any computational biology inverse problem, and its tools

should be applied as a prerequisite before attempting higher levels. Identifiability analysis should be performed before parameter estimation. The structural identifiability $\equiv$ observability equivalence provides a clean theoretical foundation that practitioners should exploit routinely.

3. Level 2: Model Discovery

3.1. Mathematical foundations

Model discovery extends Level 1 by treating the model structure itself as an unknown. Various statistical and machine learning frameworks have been explored for model discovery, including symbolic regression, grammar-based methods, sparse regression, neural networks, Gaussian process regression, and Bayesian approaches, among others (Džeroski and Todorovski, 2008; Gyori and Bachman, 2021; Brunton and Kutz, 2022; Camps-Valls et al., 2023; Métayer et al., 2026). In sparse regression methods (Mangan et al., 2016; Brunton et al., 2025), the system dynamics are represented as a sparse combination of candidate basis functions drawn from a library $\Phi = \{\phi_1, \dots, \phi_P\}$:

$$\dot{\mathbf{x}}(t) = \sum_{p=1}^P \xi_p \phi_p(\mathbf{x}(t), \mathbf{u}(t), \theta_p) \quad (3)$$

and the joint optimization problem seeks both the sparse coefficient vector ξ (defining the model structure) and the continuous parameters θ :

$$\min_{\xi, \theta} J(\xi, \theta) + \lambda \|\xi\|_q \quad (4)$$

where the regularization term $\lambda \|\xi\|_q$ penalizes model complexity, encoding Occam's razor as a formal constraint. The hyperparameter λ governs the trade-off between fidelity to data and parsimony of the model.

Excitation, exogeneity, and endogenous inputs. Model discovery, like parameter estimation, succeeds only when the data are sufficiently informative: the candidate-library columns must be linearly independent and well conditioned over the sampled region of state and input space, exactly as parameter identifiability requires a well-conditioned parameter-sensitivity matrix. A convenient way to meet this requirement is to use exogenously actuated and persistently exciting inputs. It should be noted, however, that they are neither necessary nor sufficient. For example, a poorly designed actuator is uninformative, and rich natural variation can be equally adequate. A qualitatively different difficulty arises when the inputs are not externally imposed but generated endogenously by feedback, $\mathbf{u}(t) = \pi(\mathbf{x}(t))$. The observations then explore only the closed-loop trajectories produced jointly by the dynamics and the policy, so different combinations of dynamics and feedback can reproduce the same data. Moreover, disturbances or unmeasured variables may correlate the input with the equation error and bias

the estimates. Recovering the model in that regime is a closed-loop identification problem, which requires a joint model of the input mechanism, instrumental information, or an explicit measurement-error model. This distinction becomes central at Levels 3 and 4, where the allocations are treated as endogenous behavior to be explained rather than as identification inputs.

3.2. Assessment

Supporting view

Model discovery addresses a genuine and practically important gap in computational biology: the assumption that the model structure is known is frequently untenable, and trial-and-error model building is slow and error-prone. Sparse regression approaches have demonstrated their ability to recover correct model structures from data in a range of biological settings (Kaiser et al., 2018; Mangan et al., 2016). The information-theoretic model selection criteria that accompany these approaches (AIC, BIC, Bayesian model selection) provide principled, quantitative tools for navigating the structure–complexity trade-off. The connection to AutoML and neural architecture search in machine learning (see Section 8) suggests that the community is developing increasingly powerful tools for exactly this class of problems.

Critical view

Model discovery extends parameter non-uniqueness to ambiguity between model structures: distinct mechanisms may generate identical observable dynamics. For example, different reaction networks can admit dynamically equivalent realizations (Szedekényi, 2010). Parametric identifiability within an assumed model therefore does not establish mechanistic correctness, while non-identifiability alone does not refute a mechanism. Rather, it indicates that the specified observations cannot uniquely determine its unknown parameters (Villaverde et al., 2016; Massonis et al., 2023a). In sparse regression, explicit selection of active terms introduces a combinatorial problem, whereas sparsity-promoting relaxations such as the ℓ_1 penalty are continuous but nondifferentiable at zero (Brunton and Kutz, 2022). The choice of library Φ introduces a further modeling assumption: recovery depends on whether the candidate functions adequately represent the governing dynamics. Thus, an incomplete or poorly chosen library can yield a misleading structure (Brunton et al., 2016; Mangan et al., 2016). The regularization parameter λ also influences the selected support by controlling the trade-off between fit and sparsity.

Consequently, structural conclusions should be assessed for sensitivity to regularization, noise, and library selection, rather than inferred from a single sparse fit (Fasel et al., 2022).

Synthesis

Level 2 represents a principled generalization of Level 1 that is increasingly tractable given modern computational tools. In the case of sparse regression methods, their main limitation is that it requires careful specification of the candidate library and rigorous validation of the selected structure against held-out data. Model selection at this level should be understood as providing statistical preferences, not uniqueness guarantees. The final aim is to obtain identifiable and observable models which are also mechanistically interpretable (Massonis et al., 2023a). Biological prior knowledge should be used to constrain the library and thereby reduce combinatorial non-uniqueness. Ensemble modeling can also be exploited to improve robustness, especially in information-poor scenarios (Fasel et al., 2022; Massonis et al., 2023b).

4. Level 3: Inverse Optimal Control

4.1. Mathematical foundations

Inverse optimal control (IOC) introduces the hypothesis that observed biological trajectories are not merely solutions to differential equations, but solutions to dynamic optimization problems (Banga and Sager, 2025; Albrecht et al., 2012). The framework is bilevel: an inner optimal control problem models the system’s “strategy,” and an outer inverse problem recovers the unknown objective from data.

Inner problem (optimal control):

$$\mathbf{u}^* = \arg \min_{\mathbf{u}(t)} \int_{t_0}^{t_f} \sum_{m=1}^M w_m L_m(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}) dt \quad (5)$$

subject to the system dynamics and feasibility constraints on states and controls.

Outer problem (inverse optimization):

$$\min_{\mathbf{w}, \boldsymbol{\theta}, \boldsymbol{\xi}} J_{\text{outer}}(\mathbf{w}, \boldsymbol{\theta}, \boldsymbol{\xi}) + \lambda \mathcal{R}(\mathbf{w}, \boldsymbol{\xi}) \quad (6)$$

subject to $\mathbf{u}^*$ being the solution of the inner problem, normalization of $\mathbf{w}$, and parameter bounds.

4.2. The biological motivation

The central biological claim at Level 3 is that evolution has shaped certain biological systems to behave in ways that are approximately optimal with respect to some performance criterion. This is not a new claim, as it underlies many studies in theoretical biology (Parker and Smith, 1990; Alexander, 2001), behavioral ecology (McFarland, 1977), and systems

biology (Heinrich and Schuster, 1996; Klipp et al., 2002); see (Banga and Sager, 2025) for a historical perspective. The question is whether it is sufficiently well-supported to justify the computational investment of the IOC framework.

Supporting view

The empirical evidence for optimality in biology spans a remarkable range of scales and systems (Banga and Sager, 2025). For example, at the molecular level, the native state of a folded protein minimizes free energy (Anfinsen, 1973). At the pathway level, optimal control of enzyme expression in simple metabolic networks predicted the sequential pulse-like gene expression patterns subsequently confirmed experimentally (Klipp et al., 2002; Zaslaver et al., 2004). Flux balance analysis, which assumes cells maximize growth rate subject to stoichiometric constraints, predicts the growth phenotypes of *E. coli* across a wide range of genetic and environmental perturbations (Ibarra et al., 2002; Segre et al., 2002). At the organism level, the geometry of vascular branching, the design of compound eyes, and the mechanics of human gait are all well-described by optimality models (Murray, 1926; McNeill Alexander, 1996; Mombaur and Clever, 2017).

Critical view

The evidence for optimality is real but requires careful interpretation. Several objections deserve explicit treatment. First, a long list of confirmed predictions does not establish a framework’s validity without a corresponding accounting of failures. The selectivity of evidence (presenting successes without the rate of failure) is closely related with the issues identified by Gould and Lewontin (1979) and by Schoemaker (1991). A systematic review of optimality predictions in molecular systems biology, including both confirmed and disproved cases, is conspicuously absent from the literature. Second, and most importantly, identifying the *objective function* of a biological system requires not merely that the system behaves efficiently, but that the mapping from objectives to trajectories is invertible, which the theory of IOC shows it generally is not (Banga and Sager, 2025). Objective non-identifiability is a structural feature of the IOC problem, not a numerical artifact, and it compounds all the identifiability challenges already present at Levels 1 and 2.

4.3. The identifiability challenge

Objective identifiability at Level 3 faces a new class of fundamental difficulty. The mapping from cost functionals to optimal trajectories is many-to-one: scaling the cost by any positive constant leaves the optimal trajectory unchanged. Different linear combinations of cost kernels may produce

identical optimal behaviors if their gradients are collinear along the optimal trajectory.

Recovering an objective from observed optimal behaviour requires the controls to influence the parts of the dynamics described by the candidate cost terms. If a cost term depends only on state directions that the admissible controls cannot affect, its weight cannot be recovered from behaviour alone. This reflects the classical role of controllability in optimal control (Kalman, 1960), but full controllability of the entire system is not required. Even when the relevant dynamics are controllable, the observed demonstrations must be sufficiently informative: if they explore only regions where different combinations of cost terms produce indistinguishable optimal behaviour, the corresponding weights cannot be uniquely determined (Kalman, 1964; Jin et al., 2021). At Level 3, this is assessed by asking whether the optimality conditions from all demonstrations determine a unique normalized weight vector. Exact ambiguity would be equivalent to structural non-identifiability for the available demonstrations. Similarly, poor separation between alternative weights in the presence of noise or numerical error would represent practical non-identifiability. These ideas follow the approach of Jin et al. (2021), which considers objectives as weighted feature combinations and relates recoverability to the rank of a trajectory-dependent recovery matrix.

Cross-level compensation is a further source of non-uniqueness: a change in the inferred objective $\mathbf{w}$ may be offset by a compensating change in the model parameters θ , leaving the observed output unchanged. Degeneracies with different explanations are not themselves new: Level 1 has parameter non-identifiability and Level 2 has structural indistinguishability. But in Level 3 the compensation happens between the *objective* and the *model*, i.e. the inferred purpose and inferred mechanism become jointly non-identifiable. This is further compounded when the objective and the model parameters are inferred from the same endogenous demonstrations: the controls then reflect the joint action of the dynamics and the underlying decision rule, so the recovery inherits the closed-loop identification difficulties discussed in Section 3.

Synthesis

The IOC framework is biologically motivated, mathematically coherent, and supported by a meaningful body of empirical evidence. However, it requires the following conditions to be in place before it can be applied with confidence: (i) a prior argument, independent of the framework itself, that the system of interest has been subject to sustained selection pressure on the relevant timescale; (ii) experimental designs rich enough to ensure that the observed trajectories cover a sufficiently large region of state-control space to render the objective identifiable; (iii) a strict separation between inference data and validation data, with the inferred objective tested

against new experimental conditions not used in its derivation; and (iv) explicit acknowledgment of the structural identifiability limitations of the inferred objective, including the identification of cost components that are inherently non-identifiable given the system's controllability structure. The framework should be understood as generating testable hypotheses about biological objectives, not as proving what those objectives are (Banga and Sager, 2025).

5. Level 4: Inverse Differential Games

5.1. Mathematical foundations

The inverse differential game (IDG) extends the IOC framework to multi-agent settings in which N biological agents interact strategically (Başar and Olsder, 1998; Awasthi and Lamperski, 2020; Molloy et al., 2017, 2022). Each agent n minimizes its own cost functional subject to the actions of all other agents, and the inner problem seeks a Nash equilibrium: a joint strategy $(\mathbf{u}_1^*, \dots, \mathbf{u}_N^*)$ from which no agent can unilaterally deviate to reduce its own cost.

The outer inverse problem seeks to recover the individual objective matrices $\mathbf{W} = [\mathbf{w}_1, \dots, \mathbf{w}_N]$ from observations of the joint equilibrium trajectories, together with the model structure ξ and parameters θ .

5.2. Assessment

Supporting view

The IDG framework addresses biological scenarios of genuine importance, e.g. microbial competition for shared nutrients, co-evolutionary arms races between hosts and pathogens, and competitive resource allocation in ecological communities. The suggested mathematical structure (non-cooperative differential games with Nash equilibrium inner solutions) is a natural extension of the IOC framework and is well-supported by the theory of evolutionarily stable strategies in evolutionary biology (Smith, 1978, 1982). The corresponding inverse problem in artificial intelligence (multi-agent inverse reinforcement learning) is an active research area, providing a rich set of computational tools and theoretical results that can be adapted to biological applications. For the simpler case of linear-quadratic differential games, conditions for the unique recovery of individual quadratic cost matrices from observed equilibrium trajectories have been derived under specific rank conditions on the closed-loop gains, providing at least a partial theoretical foundation (Engwerda, 2009).

Critical view

Level 4 is the most speculative component of the hierarchy. In addition to all difficulties from previous levels, it introduces: (i) the fundamental non-uniqueness of Nash equilibria, i.e. for given individual objectives, multiple self-consistent outcomes may coexist, and the data may correspond to any one of them without an independent theory of equilibrium selection; (ii) objective permutation symmetry, in which exchanging the objectives of two players with symmetric roles leaves the equilibrium unchanged; (iii) cooperative mimicry, in which the collective equilibrium behavior of N strategic agents is indistinguishable from the optimal behavior of a single aggregate agent, making the multi-agent structure undetectable from output data alone; and (iv) the non-identifiability of the number of agents N itself, which is analogous to the model selection problem at Level 2 but substantially harder. The assumption that biological agents are playing non-cooperative differential games (rather than simply responding to environmental signals through evolved regulatory programs) is a strong mechanistic claim that requires independent justification in each application. For example, bacteria competing for a carbon source are shaped by selection, not by strategic computation (West et al., 2006; Frey, 2010). Conflating these might introduce a category error that the mathematical formalism does not automatically resolve. Furthermore, the data requirements for Level 4 are severe: recovering N individual objective vectors from joint trajectory data, in the presence of all the non-uniqueness sources enumerated above, requires experimental designs of extraordinary richness that are not yet feasible in most biological systems.

Synthesis

Level 4 is best understood at present as a research agenda rather than a deployable methodology. Its mathematical foundations are sound, its biological motivation is conceptually coherent, and its connections to the machine learning literature are genuine and potentially productive. However, the theoretical conditions for reliable inference of individual agent objectives from joint trajectory data are not yet established for general nonlinear systems, and no end-to-end validation (infer objectives from data, predict new behavior, confirm experimentally) has been demonstrated in a real biological multi-agent setting. The appropriate near-term goal is to establish such demonstrations in well-controlled model systems (e.g., defined microbial co-cultures with known competitive interactions) where ground-truth

objectives can be at least partially characterized independently.

6. An illustrative case study

Our case study is inspired in previous works considering the modeling of microbial communities (Zomorodi and Maranas, 2012; Mauri et al., 2020; Díaz-Seoane et al., 2023b; Paredes-Vázquez et al., 2025). The present implementation was designed around five requirements:

1. use the same microbial system at all four levels;
2. keep the dynamics simple enough for transparent mathematical analysis;
3. introduce explicit, biologically interpretable control variables for Levels 3 and 4;
4. provide known ground truth through synthetic data; and
5. expose non-identifiability rather than forcing exact recovery.

The result is a basic proof-of-concept with simplicity and clarity in mind, not a claim that objectives can already be reliably inferred from ordinary microbial abundance data. In particular, dynamic allocation measurements or appropriate molecular proxies would be needed before applying the upper levels to a real consortium.

Figure 2 summarizes how the same microbial model supports four progressively more explanatory inverse questions, while also making explicit the change in the epistemic status of the allocation variables between the lower and upper levels. We implemented this case study using Matlab. Pseudo-codes of the simplified methods used to solve each level, and details about the Matlab implementation are given in the Appendix A. The code is available at <https://github.com/cblmbg/inverseproblemsbiocybernetics>.

6.1. Biological system and dynamic model

6.1.1. Two-strain cross-feeding consortium

Two microbial strains X_1 and X_2 compete for a common substrate S . Strain 1 secretes metabolite M_1 , which supports growth of strain 2, while strain 2 secretes metabolite M_2 , which supports growth of strain 1. The state is

$$\mathbf{x}(t) = [X_1(t) \quad X_2(t) \quad S(t) \quad M_1(t) \quad M_2(t)]^T. \quad (7)$$

The dimensionless variable $u_i(t) \in [0, 1]$ represents the effective fraction of strain i 's metabolic resources allocated to secretion. Increasing u_i increases secretion of M_i , but reduces the strain's instantaneous growth capacity. This construction creates a biologically interpretable trade-off between individual growth and support of the partner.

For Levels 1 and 2, the primary laboratory interpretation is an engineered consortium in which expression of each secretion pathway is actuated by an external input, such as an inducer concentration or an optogenetic light signal. Strictly speaking, the experimenter prescribes this actuator

Microbial community: four inverse problems

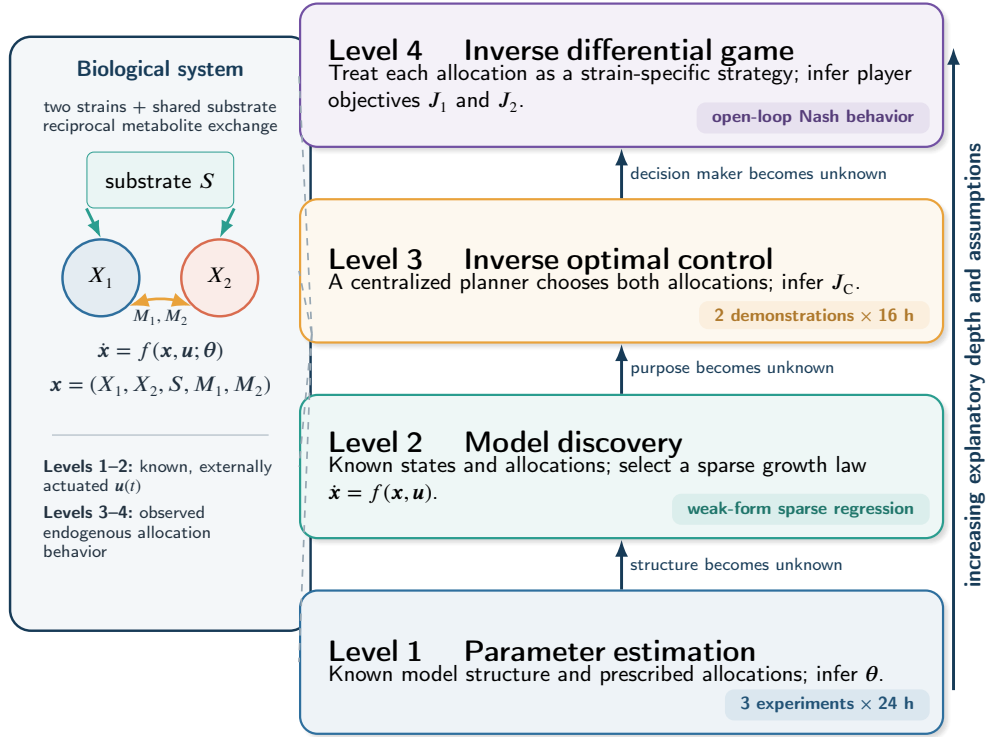


Figure 2: The four-level inverse ladder for the two-strain cross-feeding case study. Levels 1 and 2 use known, externally actuated allocations to identify parameters and dynamic structure. Levels 3 and 4 instead treat observed allocations as endogenous behavior to be explained by, respectively, a community objective or strain-specific objectives. The biological system and state equations remain fixed, but explanatory depth, assumptions, and the kind of non-uniqueness increase up the ladder.

rather than directly setting an intracellular resource fraction. The model assumes that a previously calibrated actuator-to-allocation relation maps the applied signal to a known effective input $u_i(t)$. Thus, “prescribed allocation” is shorthand for a prescribed actuator with a known effective allocation; it does not imply perfect direct control of cellular metabolism.

A different experiment could estimate $u_i(t)$ from a transcriptional reporter, flux proxy, or other molecular measurement without externally actuating the pathway. In that case $u_i(t)$ would be a measured covariate, not an externally prescribed input, and its measurement error and possible dependence on the cellular state should be included in the inverse problem. That measured-only design is not simulated here.

Throughout the case study, time is measured in hours (h), biomass concentrations X_i in grams of dry weight per litre (gDW L⁻¹), and extracellular substrate and metabolite concentrations S and M_i in millimoles per litre (mmol L⁻¹). The allocation variables u_i are dimensionless.

The status of u_i changes deliberately along the inverse ladder. In Levels 1 and 2 it is a known exogenous input used to excite and identify the dynamics. In Level 3 it becomes an observed endogenous trajectory hypothesized to solve a centralized optimal-control problem. In Level 4, u_i is instead the endogenous strategy of strain i in a differential game. The

four levels therefore share the same biological system and state equations, but the upper levels use separately generated control trajectories and ask a different question about their origin. To be clear, applying inverse optimal control or an inverse game directly to an externally imposed $u_i(t)$ could at most explain the experimenter’s actuation policy (i.e. it would not identify a biological objective of the strains).

6.1.2. Growth and material-balance equations

For strain i , let $j = 3 - i$ denote its partner. The unburdened specific growth rate is

$$\bar{\mu}_i(\mathbf{x}) = \mu_{i,\max} \frac{S}{K_{S,i} + S} \frac{M_j}{K_{M,i} + M_j}, \quad (8)$$

and the allocation-dependent growth rate is

$$\mu_i(\mathbf{x}, u_i) = \bar{\mu}_i(\mathbf{x})(1 - c_i u_i). \quad (9)$$

The implementation enforces a small positive lower bound on the allocation factor for numerical robustness. The complete dynamics are

$$\dot{X}_i = (\mu_i(\mathbf{x}, u_i) - D) X_i, \quad (10)$$

$$\dot{S} = D(S_{\text{in}} - S) - \sum_{i=1}^2 \frac{\mu_i(\mathbf{x}, u_i) X_i}{Y_{S,i}}, \quad (11)$$

Table 1

Ground-truth parameters of the synthetic microbial consortium.

Quantity	Strain 1	Strain 2	Unit
Maximum growth rate, $\mu_{i,\max}$	0.55	0.50	h^{-1}
Substrate half-saturation, $K_{S,i}$	0.80	1.00	mmol L^{-1}
Metabolite half-saturation, $K_{M,i}$	0.12	0.10	mmol L^{-1}
Allocation cost, c_i	0.35	0.30	dimensionless
Secretion rate, α_i	0.55	0.50	$\text{mmol gDW}^{-1} \text{h}^{-1}$
Substrate yield, $Y_{S,i}$	0.65	0.60	gDW mmol^{-1}
Metabolite yield, $Y_{M,i}$	8.0	8.0	gDW mmol^{-1}
Dilution rate, D	0.05		h^{-1}
Inlet substrate, S_{in}	10.0		mmol L^{-1}

$$\dot{M}_1 = \alpha_1 u_1 X_1 - \frac{\mu_2(\mathbf{x}, u_2) X_2}{Y_{M,2}} - D M_1, \quad (12)$$

$$\dot{M}_2 = \alpha_2 u_2 X_2 - \frac{\mu_1(\mathbf{x}, u_1) X_1}{Y_{M,1}} - D M_2. \quad (13)$$

Table 1 lists the ground-truth parameters used to generate the synthetic data.

6.2. Synthetic experiments and numerical implementation

6.2.1. Experimental design and excitation for the case study

The notions used here are not specific to this consortium: the distinction between structural and practical identifiability (Section 2), the requirement that the observed trajectories be sufficiently informative, and the additional difficulty created when inputs are endogenous rather than externally imposed (Section 3). This subsection states how they shape the design of the synthetic experiments.

External actuation of u_i is a convenient way to obtain informative, well-conditioned data at the lower levels, but it is neither necessary nor sufficient for identifiability. What matters is the information content of the *observed* trajectories, expressed through the observation function

$$\mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\theta}). \quad (14)$$

For Level 1, local identifiability is tied to the rank and conditioning of the stacked parameter-sensitivity matrix

$$S_{\boldsymbol{\theta}} = \begin{bmatrix} \partial \mathbf{y}(t_1) / \partial \boldsymbol{\theta} \\ \vdots \\ \partial \mathbf{y}(t_N) / \partial \boldsymbol{\theta} \end{bmatrix}. \quad (15)$$

Independent columns allow parameter effects to be separated locally, whereas poor conditioning leaves the estimates sensitive to finite sampling and noise. Replacing $\boldsymbol{\theta}$ in (15) by the augmented vector $[\boldsymbol{\theta}, \mathbf{x}]$ extends the same test to observability; here all states are measured, so this extension is not needed. For Level 2, the analogous requirement is that the regression-library columns be distinct over the sampled state-allocation region. In particular, the data should not make q_i , u_i , and $q_i u_i$ nearly collinear.

In the case study we therefore use exogenous, persistently exciting allocation schedules: they break state-allocation correlations, expand the explored operating region, and improve the conditioning of the sensitivity and library matrices. This is a design convenience, not a necessity: the allocations could instead be left endogenous while the initial conditions, inlet substrate S_{in} , and dilution rate D are varied across experiments, and it cannot rescue an intrinsically non-identifiable model. Treating an endogenous u_i as an error-free input would, however, raise the closed-loop identification issues of Section 3, which is one reason the upper-level demonstrations are generated and analyzed separately.

6.2.2. Calibration experiments

We consider three calibration experiments with different initial biomass ratios, substrate levels, and metabolite concentrations. For the synthetic engineered-consortium interpretation, the externally applied actuator signals are converted through the assumed calibrated input map into the effective allocations $u_1(t)$ and $u_2(t)$. These known inputs are persistently exciting square-wave schedules with small sinusoidal components and are imposed independently of the observed state trajectories. Across the three experiments, the realized ranges are $u_1 \in [0.130, 0.730]$ and $u_2 \in [0.150, 0.730]$. This realized variation is important because u_i affects both secretion through $\alpha_i u_i X_i$ and the growth burden through $c_i u_i$; varying it helps separate the secretion coefficients α_i from the allocation costs c_i . States are sampled at 49 times over 24 h and corrupted with independent Gaussian noise at 0.5% of the maximum state scale.

6.2.3. Upper-level demonstration experiments

Levels 3 and 4 use a separate, smaller collection of synthetic behavioral demonstrations rather than the three calibration experiments. Each upper level uses two initial states. For both initial states, the control horizon is $T = 16$ h with grid

$$t_k = 0, 2, \dots, 16 \text{ h},$$

giving eight piecewise-constant control intervals of 2 h each. Level 3 therefore generates two centralized-planner demonstrations, whereas Level 4 generates two open-loop

Nash demonstrations. The same horizon and initial states are used for the corresponding forward validations. Keeping these upper-level data separate from the calibration data is necessary because their allocations are endogenous behaviors generated by the hypothesized planner or game, rather than externally imposed identification inputs.

6.3. Level 1: parameter estimation

6.3.1. Inverse problem

The model structure in equations (10)–(13) is assumed known. The effective allocation schedules,

$$\mathbf{u}_e(t) = [u_{1,e}(t), u_{2,e}(t)]^\top,$$

are also known for each experiment e . Before estimating parameters, one should check that they are structurally identifiable. We performed this test with the STRIKE-GOLDD toolbox (Díaz-Seoane et al., 2023a), which is included within the code provided, along with the scripts that run the analyses. In this case study, if all the state variables can be measured, the 16 model parameters are indeed structurally locally identifiable. As a matter of fact, it is not necessary to measure all five states separately; it suffices to measure the substrate S , the sum of the two microbial strains, $X_1 + X_2$, and the sum of metabolites, $M_1 + M_2$, to achieve full structural local identifiability. However, if instead of $X_1 + X_2$ we measure only one of the strains (X_1 or X_2), some parameters become unidentifiable. This is the structural identifiability limit.

It should be recalled that, in practice, the estimate of a structurally identifiable parameter may be inaccurate, if the model is calibrated with scarce, noisy or insufficiently informative data. To minimize this risk, in this exercise we define a configuration that is far from the structural identifiability limit. Thus, we assume that all state variables can be independently measured, $\mathbf{y}(t) = \mathbf{x}(t)$. Furthermore, instead of attempting to estimate all 16 parameters in the model, we focus on six of them:

$$\boldsymbol{\theta} = [\mu_{1,\max}, \mu_{2,\max}, \alpha_1, \alpha_2, c_1, c_2]^\top. \quad (16)$$

All remaining parameters are treated as independently characterized.

The weighted least-squares problem is

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}_{\min} \leq \boldsymbol{\theta} \leq \boldsymbol{\theta}_{\max}} \sum_{e,k,\ell} \left[\frac{x_{e,k,\ell}^{\text{sim}}(\boldsymbol{\theta}; \mathbf{u}_e) - x_{e,k,\ell}^{\text{obs}}}{s_\ell} \right]^2, \quad (17)$$

where e , k , and ℓ index experiments, times, and states, respectively, and s_ℓ is a state-specific scale.

This is a parameter estimation problem conditional on the applied inputs. Every candidate parameter vector is simulated under exactly the same $\mathbf{u}_e(t)$. The least-squares optimizer neither estimates these schedules nor changes them to improve the fit, and Level 1 makes no claim that they are optimal choices made by the microorganisms. This distinction is what is meant by “externally prescribed” in Figure 3.

The problem is solved with a nonlinear least-squares method (Matlab’s `lsqnonlin`) from two truth-independent

Table 2

Level 1 parameter-estimation results.

Parameter	True	Estimated	Relative error (%)
$\mu_{1,\max}$	0.55000	0.55080	0.145
$\mu_{2,\max}$	0.50000	0.49919	-0.162
α_1	0.55000	0.54964	-0.065
α_2	0.50000	0.50039	0.078
c_1	0.35000	0.35175	0.501
c_2	0.30000	0.29614	-1.287

initial points: the midpoint of the parameter bounds and a seed-controlled uniform random draw within those bounds. Thus, the synthetic ground-truth vector is not used to initialize the optimizer; it is accessed only after estimation to quantify recovery error. Bound constraints enforce biologically meaningful values. It should be noted that here, for the sake of simplicity, we use `lsqnonlin`, a local solver, but that a more robust approach (needed for more complex case studies) would require the use of global optimization methods (Banga, 2008; Villaverde et al., 2022).

6.3.2. Results

The relative root mean square error across all six parameters is 0.572%. Figure 3 shows the observed and fitted trajectories for the first calibration experiment. The fitted model reproduces biomass, substrate, and metabolite dynamics while being driven by the prescribed effective allocation schedules.

If noisier data is used for calibration, the quality of parameter estimates decreases accordingly. For example, increasing the level of Gaussian noise from 0.5% to the more realistic figure of 5% yields an RMS relative error of 7.54%, which is naturally larger but may still be acceptable for many applications. Noise levels of 10% and 20% lead to errors of 16.9% and 34.8%, which would be more problematic, especially in the case of c_2 , which has the largest relative error in all scenarios. In this way we can assess how the noise level introduces a practical identifiability limit.

6.4. Level 2: model discovery

6.4.1. Candidate library

The environmental balances are retained, while the per-capita biomass growth law is treated as unknown. As in Level 1, the allocation schedules are known exogenous inputs rather than quantities discovered by the regression. This library-based sparse-regression construction follows the general data-driven model-discovery paradigm Brunton and Kutz (2022). For each strain, the candidate library is constructed from:

$$\boldsymbol{\phi} = [q_i \quad q_i u_i \quad X_1 \quad X_2 \quad S \quad M_j \quad u_i], \quad (18)$$

$$q_i = \frac{S}{K_{S,i} + S} \frac{M_j}{K_{M,i} + M_j}. \quad (19)$$

Rather than differentiating noisy biomass measurements directly, the method uses window-integrated per-capita growth:

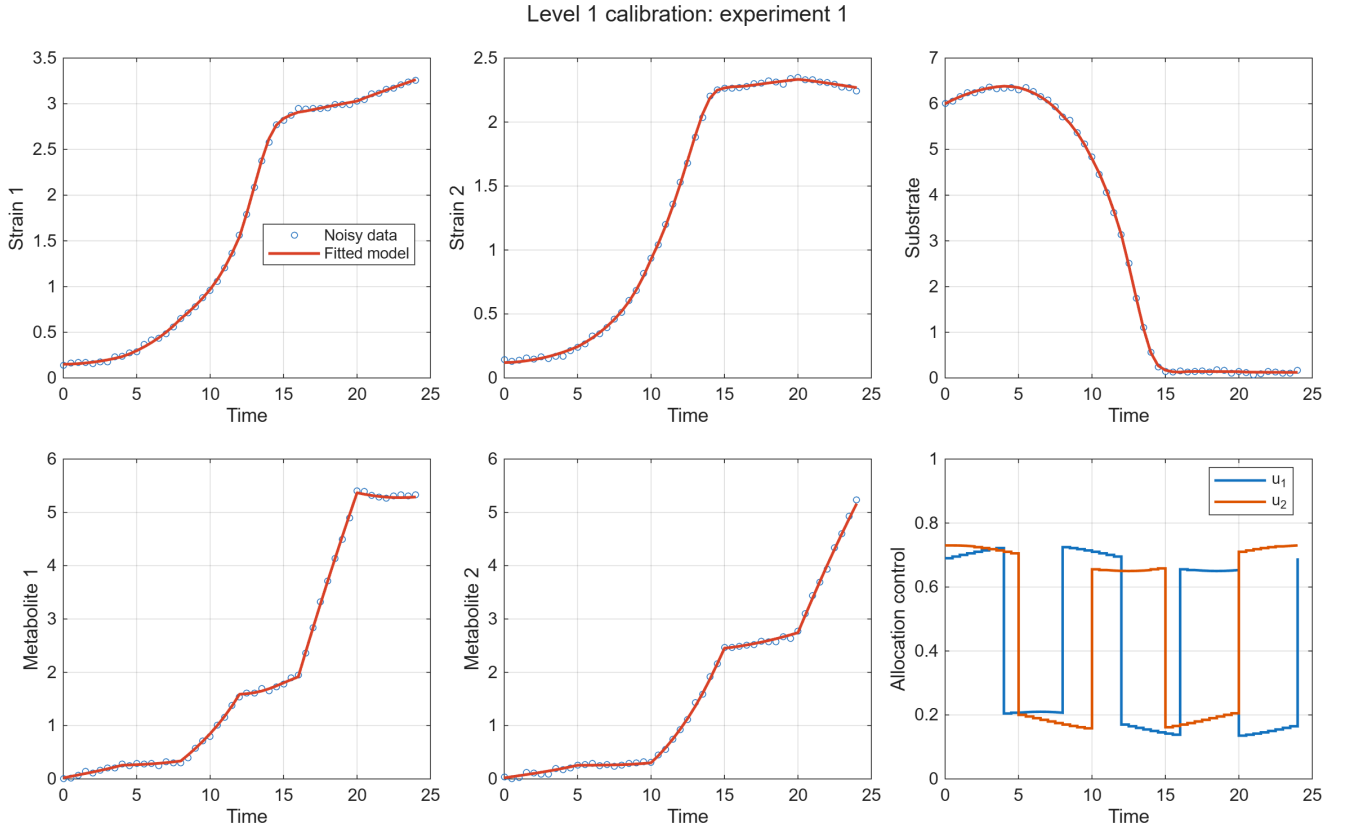


Figure 3: Level 1 calibration for the first synthetic experiment. Circles are noisy observations; solid curves are predictions obtained with the estimated parameters. The bottom-right panel shows the externally prescribed allocation inputs $u_i(t)$, interpreted as effective allocations produced by calibrated external actuation of the secretion pathways.

$$\frac{\log X_i(t_b) - \log X_i(t_a)}{t_b - t_a} \approx \beta_{i,0} + \frac{1}{t_b - t_a} \int_{t_a}^{t_b} \phi(t)^\top \beta_i dt. \quad (20)$$

Savitzky–Golay smoothing (Schafer, 2011) is applied before integration. The allocation-dependent candidates $q_i u_i$ and u_i are integrated with a left-held, zero-order-hold convention, consistent with the piecewise-constant control used to generate the data. LASSO selects a sparse coefficient vector, with the penalty chosen by leave-one-experiment-out cross-validation so that overlapping integration windows never straddle a train/validation split. Generalization is then assessed on separate held-out experiments.

This approach is closely related to weak SINDy (Messenger and Bortz, 2021). With $z_i = \log X_i$, the window equation is the integral, or constant-test-function, form of $\dot{z}_i = \beta_{i,0} + \phi^\top \beta_i$; it replaces a pointwise derivative by endpoint differences and integrals of the candidate library. General WSINDy instead uses a bank of smooth, compactly supported test functions and transfers derivatives to those functions by integration by parts. The present algorithm is therefore best viewed as a restricted weak-form SINDy method using boxcar windows, explicit smoothing, cross-validated LASSO, and post-fit thresholding.

Table 3

Level 2 sparse growth-law coefficients. Zeros denote terms removed by regularization.

Candidate term	Strain 1	Strain 2
Intercept	-0.0519	-0.0279
Monod cross-feeding, q_i	0.5177	0.4076
Allocation interaction, $q_i u_i$	-0.1887	0
Strain 1 abundance	0	0
Strain 2 abundance	0	0
Substrate	0	0
Received metabolite	0	0
Direct allocation, u_i	0	-0.0337
In-sample R^2	0.9567	0.9425
Held-out R^2	0.9430	0.9622

6.4.2. Results and structural ambiguity

For strain 1, the discovered support agrees with the generating model: growth depends on q_1 and the interaction $q_1 u_1$. For strain 2, the method selects a direct allocation term instead of $q_2 u_2$, despite achieving an in-sample $R^2 = 0.94$ and a comparable held-out $R^2 = 0.96$. This is not treated as a software failure. It is an example of structural indistinguishability caused by correlated candidate functions over the observed region of state-control space. A high predictive

Four-level inverse ladder for a microbial consortium

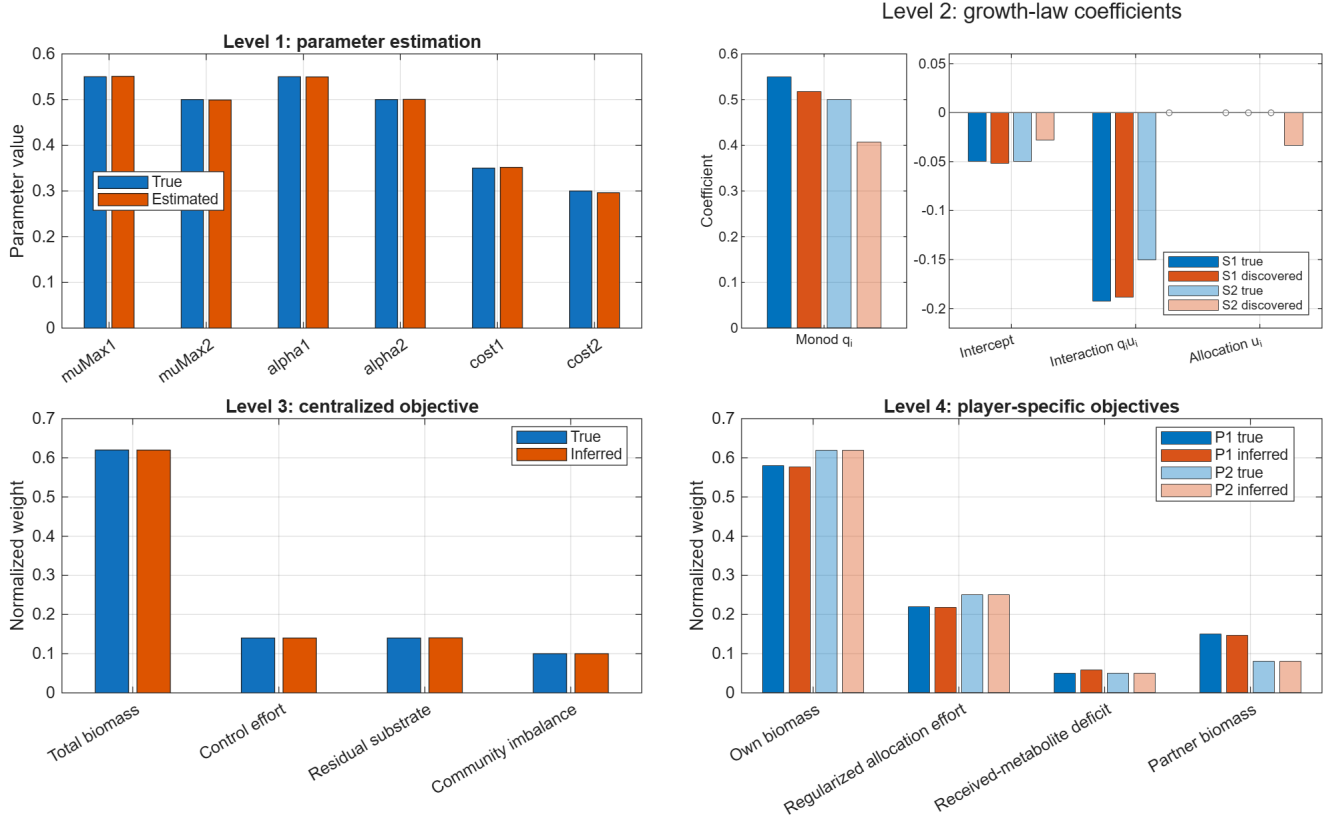


Figure 4: Summary of the four inverse levels, each comparing ground-truth and recovered quantities. Level 1 shows true versus estimated kinetic parameters; Level 2 shows the discovered growth-law coefficients against ground truth (only the terms active in the true or discovered model are shown, with the dominant Monod term on a separate scale so the smaller coefficients remain legible); Levels 3 and 4 show true versus inferred objective weights. In the Level 2 panel, S1 and S2 denote strains 1 and 2; bar colour encodes true (blue) versus discovered (orange) coefficients, while opacity distinguishes the strains (solid: S1; translucent: S2). Coefficients that are exactly zero (a candidate term absent from that model) are drawn as an open circle on the baseline. Strain 2 illustrates structural indistinguishability: its true interaction term q_{22} is dropped and a spurious direct-allocation term u_2 is selected, even though the predictive fit is good (in-sample $R^2 = 0.94$). In the Level 4 panel, P1 and P2 denote the two players (Strains 1 and 2), with the same convention (colour: true/inferred; opacity: player).

score does not guarantee recovery of the correct biological mechanism.

6.5. Level 3: inverse optimal control

6.5.1. Centralized-planner hypothesis

In contrast to Levels 1 and 2, the allocation trajectories are now endogenous: they are observations to be explained rather than actuated identification inputs. At Level 3, the consortium is provisionally treated as a single decision maker choosing both allocation trajectories. This is a hypothesis to be tested, not an assertion of community-level selection. Multilevel optimization has previously been used for forward metabolic modeling of microbial communities Zomorodi and Maranas (2012); here the objective is instead inferred from dynamic trajectories. The finite-horizon objective is linear in four normalized kernels:

$$J_C(\mathbf{u}; \mathbf{w}) = \sum_{m=1}^4 w_m F_m(\mathbf{x}, \mathbf{u}), \quad \mathbf{w} \geq 0, \quad \mathbf{1}^T \mathbf{w} = 1. \quad (21)$$

The kernels represent:

1. negative total biomass, so minimization favors biomass production;
2. quadratic allocation effort;
3. residual substrate;
4. squared biomass imbalance $(X_1 - X_2)^2$.

The forward optimal-control problem uses piecewise-constant controls on eight intervals and fourth-order Runge-Kutta integration. It is solved with a nonlinear programming solver (Matlab's `fmincon`) and box constraints $0.02 \leq u_i \leq 0.95$.

6.5.2. Relation to bilevel inverse optimal control

The general inverse optimal-control framework of Tsiantis, Balsa-Canto, and Banga Tsiantis et al. (2018) was formulated as a bilevel problem. In a simplified version with observed allocation controls and known dynamics, the structure would be

$$\hat{\mathbf{u}} = \arg \min_{\mathbf{u} \in \Delta} D(\mathbf{u}^{\text{obs}}, \mathbf{u}^*(\mathbf{w})), \quad \mathbf{u}^*(\mathbf{w}) \in \arg \min_{\mathbf{u} \in \mathcal{U}} J_C(\mathbf{u}; \mathbf{w}),$$

(22)

where $\Delta = \{\mathbf{w} \geq 0 : \mathbf{1}^\top \mathbf{w} = 1\}$, $\mathcal{U}$ includes the dynamic and control constraints, and D is a trajectory-mismatch metric. More general versions can estimate model parameters and unmeasured controls simultaneously. A direct numerical solution of (22) would call the forward optimal-control solver inside every outer evaluation of a candidate $\mathbf{w}$.

Because that nested problem is demanding, the practical strategy in Tsiantis et al. (2018) decomposed it into input and parameter reconstruction, generation of a multiobjective Pareto set of optimal controls, and selection of the Pareto solution most compatible with the data. The MATLAB case study implemented here uses neither a direct bilevel solve nor that Pareto-set decomposition. It assumes that the kinetic model has already been calibrated and that the endogenous allocation trajectories are observed. It then replaces the inner optimization problem by first-order stationarity conditions at those observed trajectories.

6.5.3. Inverse stationarity formulation

Because the objective is linear in $\mathbf{w}$, its control gradient at an observed optimum $\mathbf{u}^*$ can be written

$$\nabla_{\mathbf{u}} J_C(\mathbf{u}^*; \mathbf{w}) = [\nabla_{\mathbf{u}} F_1 \quad \cdots \quad \nabla_{\mathbf{u}} F_4] \mathbf{w} = A_C \mathbf{w}. \quad (23)$$

Finite differences provide the feature gradients A_C . Each control is classified by its box activity: an interior control must satisfy the gradient equality $\nabla_{\mathbf{u}} J_C = 0$, whereas a control at its lower or upper bound need only satisfy the corresponding one-sided Karush–Kuhn–Tucker (KKT) inequality. Splitting the rows of A_C accordingly into an interior block A_C^{eq} , a lower-bound block A_C^{lo} , and an upper-bound block A_C^{up} , the inverse problem is

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \geq 0, \mathbf{1}^\top \mathbf{w} = 1} \|A_C^{\text{eq}} \mathbf{w}\|_2^2 + \lambda \|\mathbf{w} - \mathbf{w}_0\|_2^2 \quad \text{s.t.} \quad (24)$$

$$A_C^{\text{lo}} \mathbf{w} \geq -\tau, \quad A_C^{\text{up}} \mathbf{w} \leq \tau, \quad (25)$$

with a very small regularization parameter $\lambda = 10^{-8}$, uniform reference $\mathbf{w}_0$, and a small tolerance τ that absorbs finite-difference error. Bound-active demonstrations therefore contribute information through the inequalities instead of being discarded.

Identifiability of the normalized objective is assessed from the data alone, without using the regularizer. We first solve the inverse problem using the stationarity conditions, the inequalities associated with controls at their bounds, and the requirement that the non-negative weights sum to one. We then use linear programming to find the smallest and largest possible value of each weight among all objectives that fit these conditions equally well. If every range is smaller than a prescribed numerical tolerance, the demonstrations uniquely identify the normalized objective. Otherwise, several objectives remain compatible with the data, and the regularizer selects one representative from

Table 4

Level 3 centralized objective weights.

Objective kernel	True	Inferred
Total biomass	0.6200	0.6196
Control effort	0.1400	0.1400
Residual substrate	0.1400	0.1404
Community imbalance	0.1000	0.1000

Table 5

Level 3 cold-start forward validation. RMSE values compare the re-optimized trajectories obtained with the inferred weights against the synthetic demonstrations.

Experiment	Iterations	Control RMSE	State RMSE
1	46	1.7×10^{-5}	1.3×10^{-5}
2	44	3.2×10^{-5}	1.9×10^{-5}

this set. The singular values of the equality matrix A_C^{eq} provide an additional indication of how well the weights are determined. Because multiplying all objective weights by the same positive constant does not change the optimal behaviour, one small singular value is expected (the normalization that makes the weights sum to one removes this scale ambiguity). Additional zero singular values indicate that the equality conditions alone leave further ambiguity, while additional small values indicate sensitivity to noise or numerical error. The inequalities associated with active control bounds may still resolve this ambiguity. For this reason, the final identifiability decision is based on the ranges of feasible weights rather than on the singular values alone. In summary, we check whether the observed examples of optimal behaviour point to one unique combination of objectives or whether several combinations explain them equally well. If several remain possible, the data alone cannot be used to distinguish them, and the regularizer just chooses one of those possibilities.

6.5.4. Results

The Euclidean weight error is 5.3×10^{-4} . Across the two demonstrations the stacked stationarity system has 27 interior (equality) rows and 5 bound-active rows (2 at the lower bound, 3 at the upper bound), and the data-only weight-range test confirms that the normalized objective is uniquely identified. Both cold-started forward solves converged with a positive exit flag and closely reproduced the demonstrations, but not identically; Table 5 reports the resulting nonzero discrepancies.

The singular values of the stacked stationarity matrix are approximately

$$8.23 \times 10^{-1}, \quad 2.31 \times 10^{-1}, \quad 2.2 \times 10^{-3}, \quad \text{near zero.}$$

Here the final, near-zero singular value corresponds to the objective-weight direction that best satisfies the observed stationarity conditions. The next smallest singular value represents the closest competing direction. Their ratio is

Level 3 inverse optimal control and forward validation

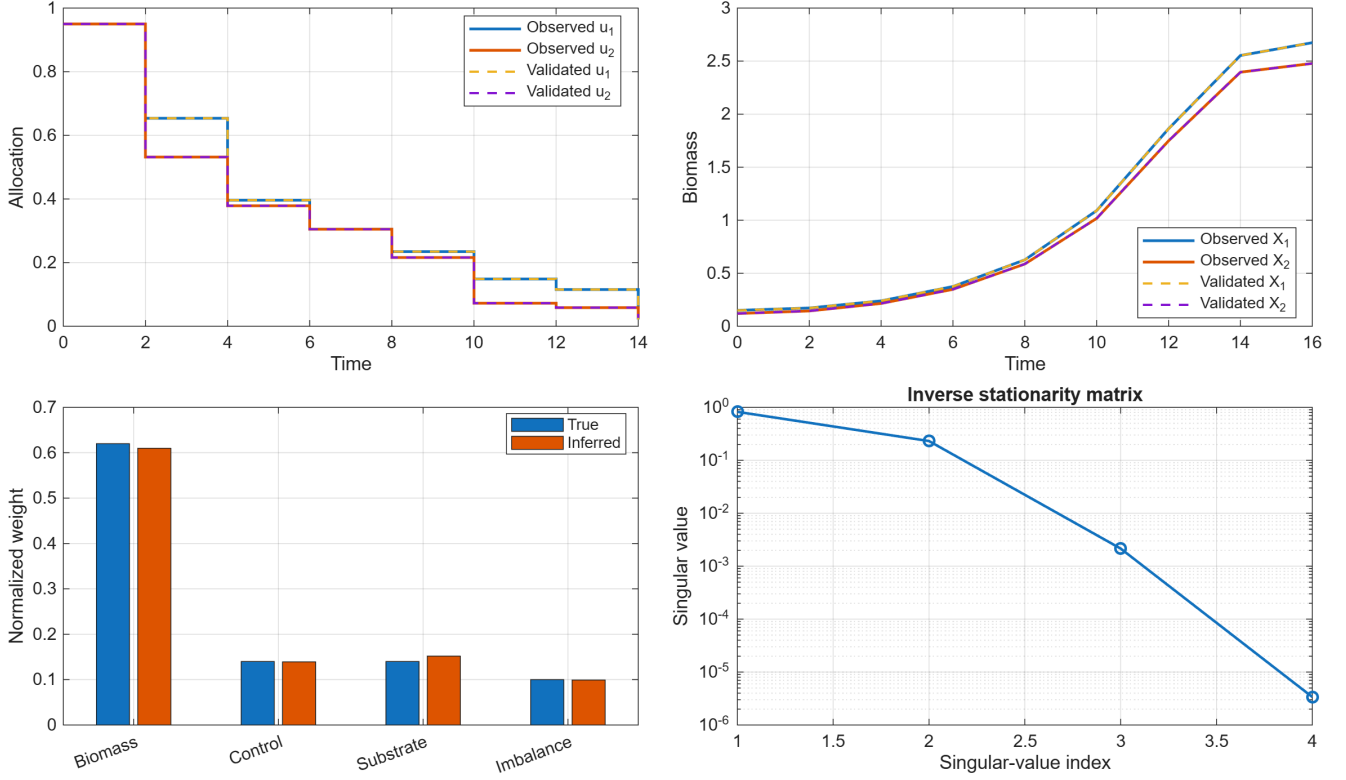


Figure 5: Level 3 results for the first experiment. Starting from the uniform default allocation $u_1 = u_2 = 0.35$, the forward solve with the inferred objective closely reproduces the observed optimal controls and biomass trajectories. The bottom-right panel shows the singular-value spectrum of the inverse stationarity matrix.

approximately 6×10^2 , so the preferred direction is well separated from that alternative and the normalized objective is locally well defined for these synthetic demonstrations. This is a local identifiability diagnostic, not proof that the same objective would remain identifiable under a different feature library, noisier measurements, or less informative trajectories.

6.6. Level 4: inverse differential game

6.6.1. Player-specific objectives

The centralized hypothesis is replaced by two independent decision makers, following the inverse noncooperative dynamic-game perspective Molloy et al. (2022). Here each $u_i(t)$ is interpreted as strain i 's endogenous strategy, rather than as an externally actuated input. Player i minimizes

$$J_i(u_i, u_j; \mathbf{w}_i) = \sum_{m=1}^4 w_{i,m} F_{i,m}, \quad \mathbf{w}_i \geq 0, \quad \mathbf{1}^\top \mathbf{w}_i = 1. \quad (26)$$

The four kernels are:

1. negative own biomass;
2. regularized own quadratic allocation effort;
3. saturating deficit of the metabolite received from the partner;

4. negative partner biomass.

More precisely, let $\bar{g} = T^{-1} \int_0^T g(t) dt$, let M_i denote the metabolite produced by player i , and let M_j denote the metabolite that it receives from its partner. The implemented features are

$$F_{i,1} = -\bar{X}_i/B_0, \quad F_{i,2} = \bar{u}_i^2 + \epsilon_M \bar{M}_i^2, \quad (27)$$

$$F_{i,3} = \frac{K_{\text{def}}}{K_{\text{def}} + M_j}, \quad F_{i,4} = -\bar{X}_j/B_0, \quad (28)$$

with $B_0 = 2$, $K_{\text{def}} = 0.35 \text{ mmol L}^{-1}$, and $\epsilon_M = 10^{-5}$ in the numerical concentration units used by the implementation. Thus, $F_{i,3}$ approaches one when the received metabolite is absent and decreases toward zero as it becomes abundant.

The $\epsilon_M \bar{M}_i^2$ contribution in $F_{i,2}$ is a numerical regularizer rather than a proposed biological expenditure. Because the produced metabolite depends on u_i , it adds weak curvature and removes otherwise flat directions when that metabolite is abundant. Across the four player-experiment demonstrations, it contributes only 0.00013%–0.00083% of $F_{i,2}$, so it does not materially change the objective values. Nevertheless, it is part of the feature basis and can affect the stationarity-matrix conditioning. The singular-value diagnostics reported below are therefore conditional on this explicitly regularized basis, consistent with the broader objective-basis-dependence caveat.

The final term permits a general-sum interaction: supporting the partner can be beneficial because the partner supplies an essential metabolite.

An open-loop Nash equilibrium (u_1^*, u_2^*) satisfies

$$J_i(u_i^*, u_j^*) \leq J_i(u_i, u_j^*) \quad \text{for every feasible unilateral deviation } u_i. \quad (29)$$

6.6.2. Forward and inverse algorithms

The forward game is solved by damped iterative best response:

1. fix u_2 and optimize u_1 ;
2. fix the updated u_1 and optimize u_2 ;
3. damp the updates and repeat until the infinity-norm change in the joint control vector is below 3×10^{-3} , subject to a 20-iteration cap;
4. verify that neither player obtains a material unilateral improvement.

Each best response is a nonlinear optimization problem (solved here using Matlab's `fmincon`). The damped control-change test is only a progress diagnostic; certification of an approximate equilibrium *requires* the unilateral-deviation check, which re-optimizes each player's control against the fixed final joint profile and demands the largest attainable objective improvement to fall below 10^{-4} , together with successful, feasible inner solves. A small damped update alone is not accepted as convergence. For the demonstrations reported here both conditions hold: the primary control-change tolerance is met after 16 of the 20 allowed iterations, with final gaps 2.15×10^{-3} and 2.49×10^{-3} and maximum unilateral improvements 2.99×10^{-7} and 4.39×10^{-7} .

The inverse problem is formed separately for each player by differentiating its four feature functionals with respect to its own control:

$$\hat{\mathbf{w}}_i = \arg \min_{\mathbf{w}_i \geq 0, \mathbf{1}^\top \mathbf{w}_i = 1} \|A_i^{\text{eq}} \mathbf{w}_i\|_2^2 + \lambda \|\mathbf{w}_i - \mathbf{w}_0\|_2^2 \quad \text{s.t.} \quad (30)$$

$$A_i^{\text{lo}} \mathbf{w}_i \geq -\tau, \quad A_i^{\text{up}} \mathbf{w}_i \leq \tau, \quad (31)$$

where, as at Level 3, each player's control rows are split into interior (equality) and lower/upper bound-active (one-sided inequality) blocks.

As at Level 3, the post-inference forward validation is deliberately cold-started from the uniform configured guess $u_1 = u_2 = 0.35$ on every control interval. The demonstrated controls are not supplied to the solver as an initial guess. Consequently, the validation tests whether damped best response can reconstruct a nearby Nash trajectory from a truth-independent initialization, rather than merely taking one local sweep from the trajectory being validated.

6.6.3. Results

The Euclidean weight errors are 1.00×10^{-2} and 1.07×10^{-3} for Players 1 and 2. Both cold-started validation solves

Table 6

Level 4 player-specific objective weights.

Objective kernel	Player 1		Player 2	
	True	Inferred	True	Inferred
Own biomass	0.5800	0.5771	0.6200	0.6191
Regularized effort	0.2200	0.2177	0.2500	0.2502
Metabolite deficit	0.0500	0.0586	0.0500	0.0504
Partner biomass	0.1500	0.1465	0.0800	0.0803

satisfy the primary control-change tolerance after 16 best-response iterations and also pass the unilateral-deviation check (Table 7).

The mean control and state RMS errors are 1.365×10^{-3} and 2.364×10^{-3} , respectively. These nonzero discrepancies are larger than their warm-start counterparts but remain small on the allocation and state scales of the case study.

The near-null-direction separation ratios are approximately 69.9 and 296 for the two players. Player 1 remains less well-conditioned, consistent with its larger weight error, particularly for the received-metabolite term.

6.7. Interpretation and limitations

What the case study demonstrates. The case study provides a simplified, but computationally coherent, realization of the inverse ladder:

- Level 1 recovers continuous kinetic parameters when the structure and controls are known.
- Level 2 admits competing structural explanations even when prediction is accurate.
- Level 3 recovers an objective only after choosing candidate kernels, normalizing their scale, and assuming centralized optimality.
- Level 4 requires an equilibrium concept, player-specific actions, and separate stationarity conditions.

The increasing data and modeling requirements are visible in the implementation. Levels 1 and 2 use ordinary time-series observations and prescribed inputs. Levels 3 and 4 additionally require controls, a candidate objective basis, and a forward optimal-control or game solver.

Assumptions and limitations. The case study presented here is a proof-of-concept example built upon several simplifying assumptions that may not accurately reflect real biological systems. Rather than stressing biological realism, its objective is to provide a clear and pedagogical illustration of the hierarchical approach introduced in the main manuscript. The main assumptions are:

1. **Synthetic ground truth.** The trajectories, objectives, and controls are generated by the same model class used for inference. This is appropriate for verification but does not establish biological validity.

Table 7

Level 4 cold-start forward validation. RMSE values compare the recomputed Nash trajectories obtained with the inferred player weights against the synthetic demonstrations.

Experiment	Iterations	Control RMSE	State RMSE	Best-response gap	Max. improvement
1	16	1.564×10^{-3}	2.509×10^{-3}	2.168×10^{-3}	3.03×10^{-7}
2	16	1.165×10^{-3}	2.220×10^{-3}	2.476×10^{-3}	4.33×10^{-7}

Level 4 inverse differential game and Nash validation

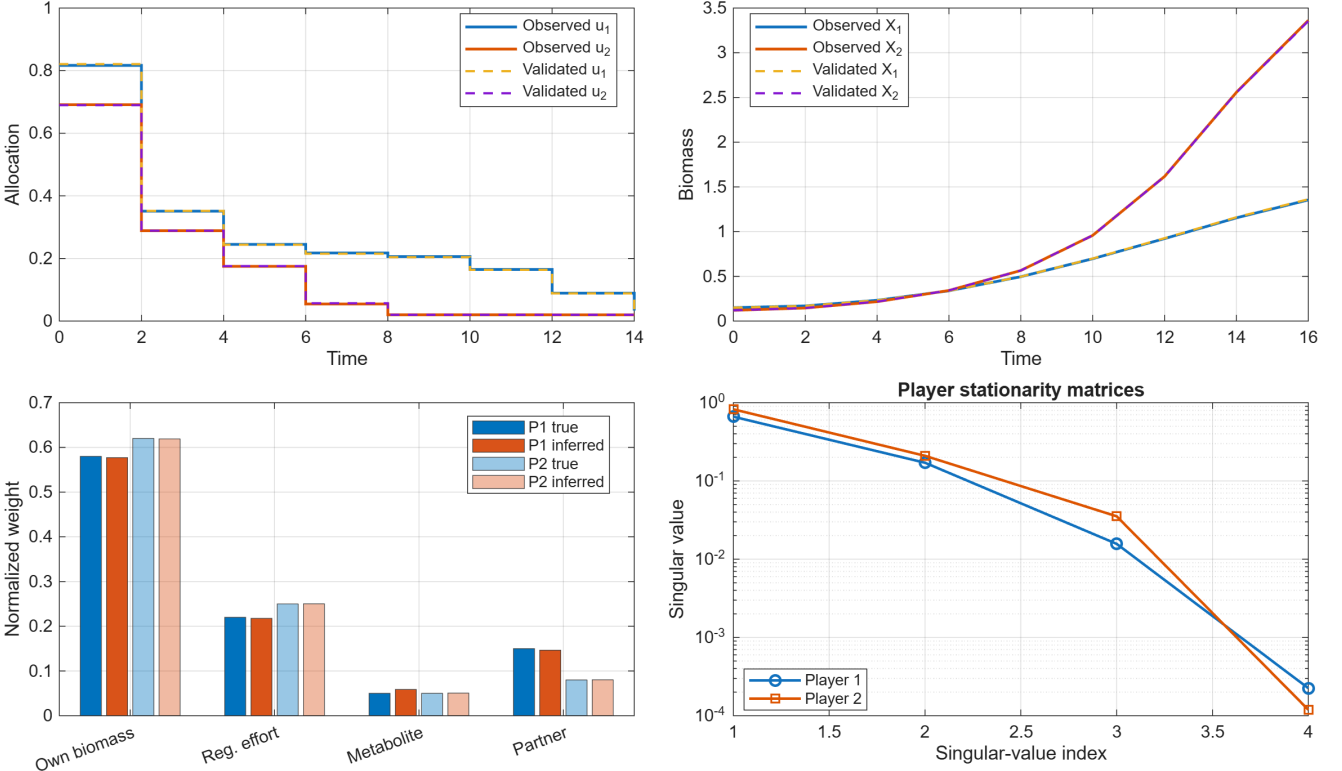


Figure 6: Level 4 results for the first experiment. Dashed cold-start forward-validation controls and trajectories, recomputed from the uniform initial guess $u_1 = u_2 = 0.35$, closely track the equilibrium data (top row). The bottom-left panel compares true and inferred player objectives: P1 and P2 denote the two players, i.e. Strain 1 and Strain 2, each carrying a four-component normalized objective-weight vector over the kernels own biomass, regularized allocation effort (Reg. effort), received-metabolite deficit (Metabolite), and partner biomass (Partner). Bar colour encodes true (blue) versus inferred (orange) weights, as in Figures 3 and 4, while opacity distinguishes the players (solid: P1/Strain 1; translucent: P2/Strain 2). The bottom-right panel shows the two players' stationarity-matrix singular-value spectra.

2. **Observed controls.** Real microbial studies rarely measure metabolic allocations directly. Transcript, protein, or secretion measurements would be required as proxies.
3. **Objective basis dependence.** Objective recovery is conditional on the selected feature library. An omitted kernel can redistribute inferred weights without visibly degrading the fit.
4. **Open-loop equilibrium.** Level 4 infers finite-horizon open-loop strategies. General nonlinear feedback Nash equilibria would require solving coupled Hamilton–Jacobi equations and are not attempted.
5. **Two-player scale.** The best-response and finite-difference methods are suitable for a compact illustration, not yet for large microbial communities.

6. **Local optimality information.** Inverse stationarity establishes compatibility with first-order conditions. Forward re-optimization and unilateral-deviation tests are therefore essential.

6.8. Conclusions

Our case study and its MATLAB implementation shows that all four levels can be represented in a single biological and computational framework. Parameter estimation and sparse model discovery are inexpensive in this simplified example, but can be tricky for larger and more realistic dynamic models. Inverse optimal control is feasible for a small controlled community when allocation trajectories are available. A two-player inverse differential game is also

computationally feasible under open-loop, finite-horizon assumptions, but it introduces a substantially larger modeling commitment and more severe conditioning issues.

The central lesson is not that the upper levels produce uniquely correct biological objectives. Rather, the implementation demonstrates how to formulate those hypotheses, diagnose their identifiability, and test their predictive consequences. The Level 2 structural ambiguity and the Level 4 conditioning difference between players are therefore significant results: they make the case study consistent with the manuscript's emphasis on both opportunities and limitations.

7. The Broader Question: Are Optimality Principles Appropriate for Biology?

The upper two levels of the hierarchy rest on the biological claim that evolution produces systems that are, at least approximately, optimal with respect to some identifiable objective. This claim has a long and contested history, and its assessment has direct consequences for the practical utility of Levels 3 and 4. Below we provide a condensed view, more details can be found in (Banga and Sager, 2025).

7.1. Historical context and the Gould–Lewontin critique

The application of optimality principles to biology dates at least to Varro's conjecture about honeycomb geometry (Mackenzie, 1999) and received formal treatment in the mid-twentieth century through the work of Rashevsky, Rosen, and others (Rosen, 1967). The optimality research program in evolutionary biology reached its high-water mark in the 1970s and 1980s, when behavioral ecology and life-history theory generated a substantial body of confirmed predictions using optimization models of animal behavior, foraging strategies, and reproductive allocation (Parker and Smith, 1990; Pyke et al., 1977).

The same period produced the most serious critique of the program. Gould and Lewontin's "Spandrels of San Marco" (Gould and Lewontin, 1979) argued that the adaptationist program (the tendency to explain every biological trait as an optimal adaptation to some function) was structured to be immune to refutation. When a predicted optimum failed to match observation, the modeler could always escape by redefining the currency being optimized, relaxing the assumed constraints, or appealing to historical contingency. The framework was, in this reading, not a scientific theory but a post-hoc rationalization engine. This critique remains influential and cannot be dismissed. However, it is important to understand precisely what it established. The spandrels argument is a critique of *unconstrained* adaptationism, the tendency to freely choose both the objective and the constraints until the data are explained. It is not a demonstration that optimality principles are never applicable in biology. A substantial community of evolutionary and systems biologists defended the use of optimality reasoning while conceding that constraints must be specified independently

of the data being explained (Smith, 1978; Parker and Smith, 1990; Alexander, 2001; Sutherland, 2005).

7.2. The proof versus usefulness distinction

A productive resolution of this long-standing debate comes from distinguishing between two questions that are frequently conflated:

1. Can optimality principles in biology be *proven correct* in the way that, say, the principle of least action is derived in classical mechanics from the Hamiltonian formalism?
2. Are optimality principles in biology *useful* in the scientific sense: do they generate novel hypotheses, motivate new experiments, and organize observations into coherent mechanistic accounts?

The answer to the first question is no, and this concession should be made explicitly. The principle of least action in physics rests on a mathematical structure (the Hamiltonian) that is exact, universal within its domain, and derivable from first principles. Nothing in biology enjoys this status. Biological optimality is always approximate, always domain-specific, and always conditional on constraints that must themselves be characterized empirically. To demand the same epistemological standard of biological optimality modeling as of Hamiltonian mechanics is to set a bar that physics itself rarely meets at the research frontier.

The answer to the second question is yes, with important qualifications. Scientific frameworks do not need to be fully correct to be productive. Newtonian mechanics is "wrong" in the sense that it breaks down at relativistic speeds; the ideal gas law is "wrong" in the sense that real gases deviate from it; the Hardy-Weinberg principle is "wrong" in the sense that real populations are subject to selection, drift, and migration. None of these frameworks has been abandoned, because they are useful: they organize observations, generate predictions, and serve as baseline models against which more complex theories are assessed.

Optimality modeling in biology satisfies the usefulness criterion in the same sense. It generates hypotheses (e.g. about which metabolic objective a pathway is maximizing, about what constraint structure explains a regulatory pattern) that can be tested experimentally. When those predictions are confirmed (as in the gene expression pulse case), the framework has earned its place. When they fail (as they sometimes do), the failure is itself informative, pointing toward missing constraints or misspecified objectives. The appropriate attitude toward biological optimality is neither uncritical acceptance nor blanket rejection, but the same empirical scrutiny that any scientific framework deserves.

7.3. Conditions for responsible application

The "useful rather than correct" framing does not, however, constitute a free pass. Usefulness must be falsifiable to be scientifically meaningful. Several conditions should govern the responsible application of optimality modeling in computational biology:

1. **Prior biological justification.** An argument, independent of the data being modeled, that the system of interest has been subject to sustained and directional selection pressure on the relevant timescale. Systems that have evolved in stable environments over many generations under strong selection (e.g., central metabolic pathways of model organisms) are better candidates than recently diverged systems or those under relaxed selection.
2. **Pre-specified constraints.** The constraints of the optimization problem should be derived from independent biological knowledge (e.g. physical limitations, resource budgets, known regulatory architectures), not fitted to the data. The Gould-Lewontin critique has full force against post-hoc constraint specification.
3. **Prospective validation.** Inferred optimality principles should be used to make predictions about system behavior under new experimental conditions not used in the inference step. Retrospective fit to training data is insufficient.
4. **Explicit treatment of identifiability.** The structural identifiability of the inferred objective must be assessed, and non-identifiable components must be reported rather than suppressed.
5. **Honest reporting of failures.** Instances where optimality predictions failed should be reported alongside successes. The scientific community cannot assess the base rate of success of the framework without access to both.

These conditions are demanding. They might rule out a fraction of published optimality modeling in biology, which does not meet all of them. But they are the conditions under which optimality modeling earns its place as rigorous science rather than sophisticated curve-fitting.

8. Connections to Machine Learning and Artificial Intelligence

A striking feature of the four-level hierarchy is that each level has a close mathematical relative in modern machine learning and artificial intelligence, and those relatives have been deployed at scale with considerable success. This parallel deserves careful examination, as it provides both encouragement and important cautions for the biological application. Figure 7 summarizes the correspondence between the four levels of the biological inverse problem hierarchy and their machine learning analogues.

8.1. What the correspondence supports

The machine learning evidence provides genuine encouragement for the biological hierarchy. Neural network training demonstrates that high-dimensional, non-convex parameter estimation is tractable at massive scale with modern optimization methods. AutoML and neural architecture search demonstrate that simultaneous structure and parameter identification is achievable and practically useful (He

et al., 2021; Ren et al., 2021). Inverse reinforcement learning (IRL) (Ng and Russell, 2000; Abbeel and Ng, 2004; Arora and Doshi, 2021) demonstrates that inferring reward functions from observed behavior is a well-posed and tractable problem in structured settings, and underlies much of the recent progress in learning from human demonstration in robotics and autonomous systems. Multi-agent IRL (Deshpande et al., 2025) demonstrates that the Level 4 problem is being actively worked on by a large community with real progress on benchmark problems.

Crucially, the machine learning analogy is not merely a structural observation. The computational tools developed for these problems (scalable optimization algorithms, regularization techniques for sparse recovery, variational methods for IRL, game-theoretic solution concepts for multi-agent settings) might be transferable and/or synergistic to the biological domain. The biological application is the frontier, since the computational foundations are not speculative.

8.2. The disanalogies that matter

However, the correspondence has significant limits, and the biological application cannot simply inherit the ML success story without confronting them.

Data regime. Neural networks are trained on datasets of millions to billions of examples. IRL in robotics typically requires thousands of demonstrated trajectories. Biological time-series experiments (even well-funded ones) produce tens to hundreds of observations per condition, from systems with hundreds to thousands of interacting variables. The favorable scaling properties of ML methods are not incidental: they are essential to why those methods work. Importing the framework without the data is, at best, a research challenge; at worst, it produces misleading results.

Goals of inference. In ML applications, IRL is used primarily for behavioral mimicry: the goal is to build a system that acts like the demonstrated agent, and whether the inferred reward function corresponds to the agent’s “true” objective is largely irrelevant. In biology, the goal is often mechanistic inference: the inferred optimality principle is supposed to tell us something real about the organism’s evolutionary history and regulatory architecture. This difference in epistemological goals means that the ML success story might not transfer in a straightforward way. A model that mimics observed trajectories accurately but for the wrong mechanistic reasons is a useful engineering tool but gives a misleading scientific claim.

State of multi-agent IRL. Multi-agent IRL is not a solved problem in machine learning (Arora and Doshi, 2021; Deshpande et al., 2025). It works well in low-dimensional, fully observable, stationary environments with small numbers of agents. It faces serious open challenges in high-dimensional, partially observable, non-stationary settings, precisely the features of real biological systems.

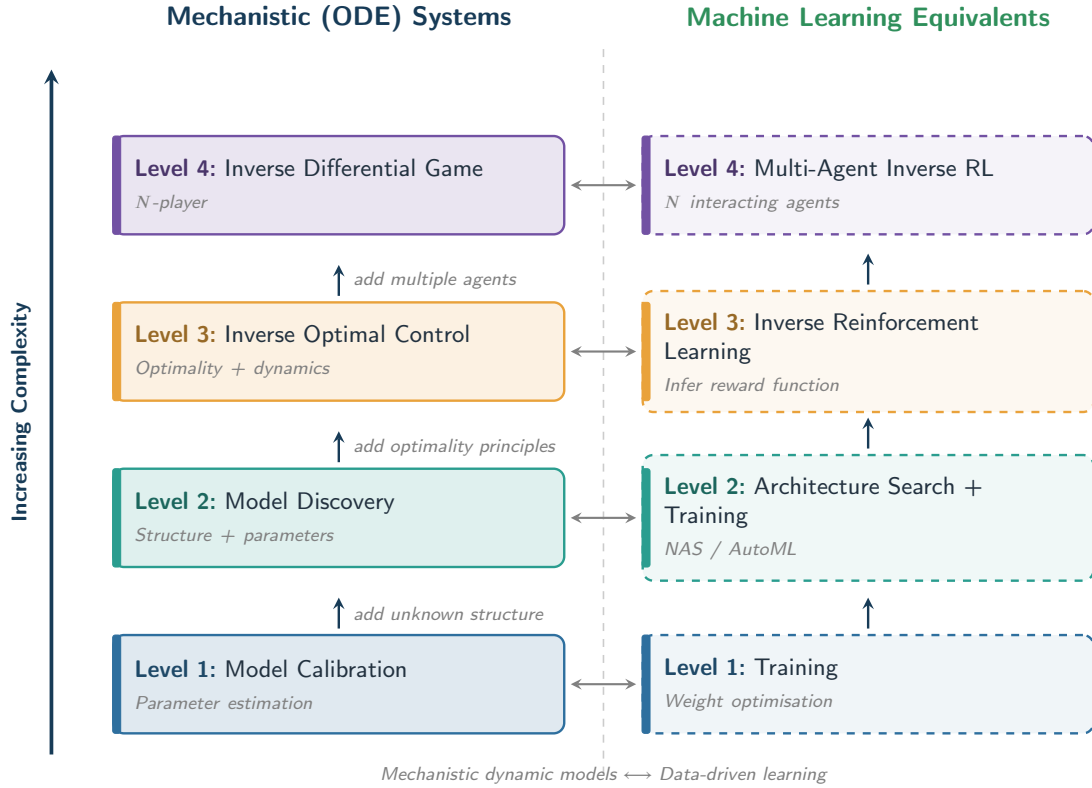


Figure 7: The hierarchy of inverse problems in systems biology (left column) and their machine learning equivalents (right column). Each level inherits the challenges of the levels below and introduces new sources of ill-posedness and computational difficulty. The horizontal arrows indicate mathematical equivalences that this study exploits to develop unified methodological frameworks.

Synthesis

The ML correspondence is genuinely encouraging and practically useful: the computational tools being developed for IRL and multi-agent IRL are natural candidates for adaptation to biological inverse problems. However, the biological application faces data and goal-of-inference challenges that have no direct counterpart in the ML settings where these methods succeed. Closing these gaps is a concrete and tractable research agenda, but it requires explicit methodological work, not the assumption that ML success implies biological tractability.

9. Conclusions

We have proposed and examined a four-level hierarchy of inverse problems in computational systems biology from both supporting and critical perspectives. Table 8 provides a unified comparison of the four levels across several dimensions relevant to their practical application in biological systems.

The table makes explicit an asymmetry that runs through the entire discussion: the lower half of the hierarchy (Levels 1 and 2) is substantially better established (theoretically, empirically, and computationally) than the upper half (Levels 3 and 4). This asymmetry has important implications

for how the framework should be presented and applied. Describing all four levels as components of a unified, deployable methodology overstates the case for the upper levels. A more honest characterization is that Levels 1 and 2 constitute established practice, while Levels 3 and 4 constitute a research frontier with clear motivation, coherent mathematics, and substantial open questions.

The following conclusions emerge from this assessment.

On Levels 1 and 2. Parametric identification and model discovery rest on rather mature theoretical and computational foundations. The tools for identifiability analysis, global optimization, and model selection are quite well developed and should be part of the standard toolkit for computational biology. In our view, these levels are appropriate for immediate application, provided that identifiability analysis is performed as a prerequisite and that model structures are validated against held-out data.

On Levels 3 and 4. Inverse optimal control and inverse differential games are biologically motivated and mathematically coherent, but they face identifiability challenges that are substantially more severe than those at Levels 1 and 2, and their biological validity depends on assumptions about evolutionary optimization that must be established empirically for each application. They should be approached as hypotheses-generating frameworks rather than inference engines, and any inferred objective or equilibrium structure should be subjected to prospective experimental validation.

Table 8

Comparative assessment of the four levels of the biological inverse problem hierarchy.

Dimension	Level 1	Level 2	Level 3	Level 4
Theoretical maturity	High	Moderate–high	Moderate	Low
Biological evidence base	Extensive	Extensive	Substantial	Limited
Key non-uniqueness[†]	Parameter non-identifiability	Structural indistinguishability	Objective non-identifiability <i>and</i> objective–parameter compensation	Equilibrium multiplicity <i>and</i> objective–parameter compensation
Key control concept	Observability	Observability (I/O equiv.)	Controllability and reachability	Joint reachability, competitive excitation
Data requirements	Moderate	Moderate–high	High	Very high
Computational burden	Moderate	High	Very high	Extreme
ML analogue maturity	High	High	Moderate	Low
Validation status	Well validated	Validated in case studies	Partial prospective validation	No end-to-end biological validation

[†] Non-uniqueness is cumulative: each level inherits the ambiguities of the levels below and adds the source named here. Levels 3 and 4 additionally admit *cross-level* compensation, in which the inferred objective trades against the model parameters (or structure), so that purpose and mechanism become jointly non-identifiable.

On optimality principles in biology. We share the view that the appropriate standard for biological optimality modeling is usefulness, not proof of correctness. Optimality principles cannot be derived from first principles in biology as they can in Hamiltonian mechanics. They can, however, generate precise and falsifiable predictions that have been confirmed in a meaningful number of cases. The conditions for responsible application (independent biological justification, pre-specified constraints, prospective validation, explicit identifiability analysis, and honest reporting of failures) are demanding but achievable.

On the machine learning connection. The formal correspondence between the four biological levels and the analogous ML hierarchy (neural network training, AutoML, inverse reinforcement learning, multi-agent IRL) is genuine and practically useful. It provides computational tools, theoretical frameworks, and existence proofs that the problems are tractable in structured settings. However, the biological application faces data scarcity, non-stationarity, and mechanistic inference goals that create a substantial gap between ML success and biological applicability. Closing this gap is a concrete and well-defined research agenda.

Future directions. The most urgent priorities are: (i) end-to-end prospective validation of the Level 3 IOC pipeline in a real biological system, demonstrating that an inferred objective can predict system behavior under qualitatively

new experimental conditions; (ii) development of identifiability theory for the joint $(\mathbf{w}, \xi, \theta)$ problem at Level 3; (iii) benchmark studies in defined biosystems (e.g. microbial co-culture systems) to assess the feasibility of Level 4 IDG inference; and (iv) systematic reporting of both successful and failed optimality predictions to allow the community to assess the base rate of success of the framework across biological domains.

In our view, the ladder of inverse problems offers a coherent and ambitious vision, not only for traditional systems biology, but also for new areas such as cybergenetics, which combines synthetic biology with digital and/or biological feedback controllers to identify and regulate cellular processes (Briat and Khammash, 2023; Caringella et al., 2023). Climbing it responsibly requires matching the level of methodological rigor to the level of biological certainty and, at every rung, about how much of the structure above has been built and how much remains to be constructed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All code required to reproduce the results of this paper is available at <https://github.com/cblmbg/inverseproblemsbio> cybernetics (tagged release v1.0.1), with a permanent snapshot deposited at Zenodo (DOI: 10.5281/zenodo.22671618), which also contains supplementary material providing: the detailed mathematical formulations used throughout, the numerical procedures applied to the case study, and a discussion on the uniqueness of solutions across the four levels.

Declaration of generative AI and AI-assisted technologies

During the preparation of this manuscript, Claude (Anthropic) was used solely to improve the language, clarity and readability of author-written text. Claude was also used for code unit-testing and refactoring. All output was reviewed and edited by the authors, who take full responsibility for the content of the publication.

A. Appendix

A.1. Pseudocodes

The four listings below summarize the algorithms used for each level of the case study. They are deliberately simplified for readability and omit implementation details (solver options, smoothing windows, tolerances), which are given in the accompanying code. A common notation is used: u denotes the allocation controls, θ the kinetic parameters, and w the normalized objective weights on the simplex $\{w \geq 0, \mathbf{1}^T w = 1\}$.

The two lower levels use the allocations as *known* exogenous inputs: Level 1 (Section 2) estimates θ by multi-start nonlinear least squares, and Level 2 (Section 3) discovers a sparse growth law by cross-validated LASSO on window-integrated per-capita growth. The two upper levels instead treat the allocations as *endogenous* optimal behavior and recover the objective from first-order (KKT) stationarity: for each demonstration the control rows are classified as interior (an equality condition, stacked into A^{eq}) or bound-active (a one-sided inequality, stacked into A^{lo} or A^{up}), and `inferSimplexWeights` solves the resulting constrained problem for w . Level 3 (Section 4) does this for a single centralized planner, and Level 4 (Section 5) player by player for an open-loop Nash equilibrium.

In every case the inferred quantities are checked by an independent forward solve that is cold-started from a uniform allocation $u = 0.35$. In this way we show that trajectory reproduction is a genuine validation rather than a warm-start consistency check.

Level 1

```
Input: experiments {t_e, xobs_e, u_e}, parameter bounds, initial guesses
for each initial guess theta0
  theta <- lsqnonlin(residual, theta0, bounds)
  residual(theta):
    for each experiment e
      xsim_e <- simulateControlledModel(theta, x0_e, t_e, u_e)
      append scaled(xsim_e - xobs_e)
select theta_hat with the smallest residual norm
Output: theta_hat and fitted state trajectories
```

Level 2

```
Input: calibration experiments, held-out experiments, Level 1 estimates
for each strain i
  smooth the measured state trajectories
  for each admissible time window
    compute window-averaged per-capita growth y
    compute averaged library row [q_i, q_i*u_i, X1, X2, S, Mj, u_i]
  (q_i*u_i and u_i integrated with a left-held zero-order hold)
  beta_i <- lasso with leave-one-experiment-out CV penalty selection
  threshold small coefficients; compute in-sample and held-out R2
  compare selected support with the expected support
Output: sparse support, coefficients, and in-sample/held-out fit
```

Level 3

```
Input: model, initial states, candidate community features
for each of 2 synthetic experiments e
  demo_e <- solveCommunityPlanner(model, x0_e, trueWeights)
  A_e <- finiteDifferenceFeatureJacobian(demo_e,
    communityCostFeatures)
  classify each control row as interior / lower-active / upper-active
  stack interior rows into A_eq, lower rows into A_lo, upper rows
  into A_up
  w_hat <- inferSimplexWeights(A_eq, LowerBoundMatrix=A_lo,
    UpperBoundMatrix=A_up)
for each experiment e
  validation_e <- solveCommunityPlanner(model, x0_e, w_hat,
    initial controls = 0.35)
Output: inferred weights, identifiability ranges, and validation errors
```

Level 4

```
Input: model, initial states, candidate features for each player
for each of 2 synthetic experiments e
  demo_e <- solveOpenLoopNash(model, x0_e, truePlayerWeights)
  for each player i
    A_i,e <- finite-difference own features with respect to own control
    classify rows as interior / lower-active / upper-active
    stack into A_eq_i, A_lo_i, A_up_i
  for each player i
    w_hat_i <- inferSimplexWeights(A_eq_i, LowerBoundMatrix=A_lo_i,
      UpperBoundMatrix=A_up_i)
for each experiment e
  validation_e <- solveOpenLoopNash(model, x0_e, w_hat,
    initial controls = 0.35)
  verify control/state errors and unilateral-improvement gaps
Output: player-specific weights and Nash validation diagnostics
```


A.2. Implementation details

To implement the case study, we used MATLAB R2026a with the Optimization, Statistics and Machine Learning, and Signal Processing Toolboxes. We also used the Symbolic Math Toolbox required by the STRIKE-GOLDD structural-identifiability analysis.

Our MATLAB implementation contains two complementary integrators that solve the same biological ODEs:

- `simulateCommunity.m` uses `ode15s` with interpolated controls and non-negativity constraints;
- `simulateControlledModel.m` uses fixed-step fourth-order Runge–Kutta integration for the repeated simulations required by optimization.

The adaptive `ode15s` solver is used as the reference integrator for generating or validating experimental trajectories. Its variable step size and error control are appropriate when accuracy and robustness to possible stiffness are more important than a fixed computational cost. It also supports continuous interpolation of measured control profiles and direct enforcement of state non-negativity.

The fixed-step Runge–Kutta solver is used inside the parameter-estimation, optimal-control, finite-difference, and best-response loops. These algorithms may evaluate the model thousands of times. A prescribed integration grid gives each evaluation the same number of right-hand-side

calls, aligns exactly with the piecewise-constant control parameterization, and avoids small changes in adaptive step selection appearing as numerical irregularities in objective functions or finite-difference gradients. For this low-dimensional model, it is therefore substantially faster and more reproducible within optimization.

The two modes are not different and therefore are not expected to produce different biological conclusions. The fixed RK4 step was chosen only after a convergence check against `ode15s`; the unit tests require their all-state RMS difference to remain below 2×10^{-3} on a representative experiment. If the model were made substantially stiffer, this comparison would need to be repeated and the optimization integrator or step size revised.

The entire workflow is launched by

```
results = runInverseLadderCaseStudy;
```

The command generates data, executes the four levels, performs forward validation, saves `inverse_ladder_results.mat`, and exports the summary graphics.

CRedit authorship contribution statement

Julio R. Banga: Conceptualization of this study, Writing - Original draft preparation, Methodology, Software. **Alejandro F. Villaverde:** Writing - Original draft preparation, Software.

Table 9
Principal MATLAB components.

File	Purpose
runInverseLadderCaseStudy.m	Top-level reproducible workflow.
defaultInverseLadderConfig.m	Parameters, experiments, objectives, and solver settings.
runLevel1aprioriIdentifiability.m	Full-state structural identifiability and observability via STRIKE-GOLDD.
runStructuralIdentifiabilityComparison.m	Structural analysis across the three observation scenarios.
runStrikeGolddAnalyses.m	Shared STRIKE-GOLDD runner.
communityRhs.m	Five-state controlled microbial ODE.
generateCalibrationData.m	Synthetic, persistently excited calibration experiments.
simulateCommunity.m	Adaptive ode15s simulator.
simulateControlledModel.m	Fixed-step RK4 simulator used in optimization.
runLevel1ParameterEstimation.m	Bound-constrained multi-experiment calibration.
runLevel2ModelDiscovery.m	Integrated sparse growth-law discovery.
communityCostFeatures.m	Centralized objective candidate kernels.
solveCommunityPlanner.m	Centralized forward optimal-control solver.
runLevel3InverseOptimalControl.m	Level 3 inverse KKT and validation.
playerCostFeatures.m	Player-specific objective candidate kernels.
solveOpenLoopNash.m	Damped iterative best-response solver.
runLevel4InverseDifferentialGame.m	Player-wise inverse KKT and Nash validation.
finiteDifferenceFeatureJacobian.m	Finite-difference feature gradients.
inferSimplexWeights.m	Bound-aware KKT recovery of simplex objective weights.
plotInverseLadderResults.m	Four-panel summary figure.
testInverseLadderCaseStudy.m	MATLAB unit tests.
kktBoundHandlingTest.m	Bound-aware inverse-KKT unit tests.

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